

MUHAMMED SHAMEEL

Data Scientist | Machine Learning Engineer

✉ muhammedshameel3009@gmail.com ☎ +91 8891232625 📍 Abu Dhabi, United Arab Emirates

🌐 linkedin.com/in/muhammed-shameel 🐙 github.com/Muhammed-Shameel

🔗 muhammed-shameel.github.io 📅 14-10-2026 📄 Visiting visa 🇮🇳 Indian

Data Scientist and MSc graduate with a BSc in Mathematics, specializing in Machine Learning, RAG applications, and FastAPI development. Hands-on experience building predictive models, data processing pipelines, and API services deployed via Docker and cloud platforms.

TECHNICAL EXPERTISE

Programming Language & Core

Python, SQL, Linear Algebra, Probability & Statistics, Hypothesis Testing, Regression

Machine Learning & Analytics

Scikit-learn, TensorFlow, Keras, PyTorch, Random Forest, XGBoost, Model Evaluation, Hyperparameter Tuning

AI & LLM Architecture

RAG, LangGraph, Vector Embeddings, Semantic Chunking, Metadata Engineering, Prompt Engineering

Backend & Cloud Tools

FastAPI, Docker, AWS Fundamentals, Git, Matplotlib, Seaborn, Tableau

WORK EXPERIENCE

InfoCreon Solutions Private Limited

Agentic AI Intern

07/2026 – 09/2026
Kerala, India

- Architected Agentic AI workflows using LangGraph and Python to coordinate multi-agent tasks, manage state transitions, and execute RAG pipelines.
- Built and served backend API microservices using FastAPI to integrate LLM decision logic with application data pipelines
- Containerized AI applications with Docker and deployed services across cloud platforms using Git, Render, and Vercel.

Cloud & Data Visualization Engineering Intern

05/2026 – 07/2026
Kerala, India

- Developed backend Python services and engineered targeted prompts to optimize LLM outputs and data processing workflows.
- Preprocessed and engineered features on Kaggle datasets to train, evaluate, and fine-tune machine learning models.
- Built FastAPI endpoints to serve model predictions and deployed containerized services using Docker, Render, and Vercel.

Mathematics Tutor

07/2023 – 03/2024

- Delivered personalized one-on-one mathematics instruction to high-school students.
- Translated complex mathematical concepts into clear, structured problem-solving frameworks to improve student understanding.

PROJECTS

Grocery AI— Multimodal RAG Knowledge Assistant

- Orchestrated a grounding-first RAG system through specialized agents for security, vision, retrieval, grounding validation, and LLM reasoning.
- Established a strict grounding gate using cosine-similarity relevance scoring to block unsupported LLM responses when retrieved evidence is insufficient.
- Decoupled embedding providers through isolated ChromaDB vector spaces, while introducing Voyage AI embeddings with checkpointed, rate-limit-aware, resumable ingestion.
- Bridged computer vision with retrieval by integrating a PyTorch CNN pipeline that dynamically anchors knowledge retrieval based on visual classification.

Gym Intelligence Platform – Agentic AI Platform

07/2026 – 09/2026

- Architected stateful multi-agent workflows with LangGraph to synthesize member, trainer, and operational data into actionable business reports.
- Made FastAPI backend endpoints combining rule-based business logic with LLM decision reasoning for operational insights.

FLUVIO-AI-Powered Flood Intelligence Dashboard

05/2026 – 07/2026

- A flood risk intelligence dashboard delivering interactive geospatial visualizations across 86 monitored regions.
- Engineered data preprocessing and feature extraction pipelines from environmental and geospatial datasets to improve predictive model accuracy.

CSIRO Image2Biomass Prediction

- Formulated regression models utilizing MSE, Huber, and LogCosh loss functions to estimate plant biomass from spectral imagery.
- Constructed a weighted ensemble pipeline to stabilize prediction accuracy across heterogeneous dataset distributions.
- Achieved $R^2 = 0.86$ on locked-in cross-validation splits using hyperparameter optimization and custom model blending.

Data-Driven Analysis on Football Players Under Different Managers

01/2025 – 09/2025

- Analyzed 2,500+ Premier League match records using Factor Analysis and multivariate feature engineering to uncover latent player performance metrics and evaluate tactical compatibility and performance adaptation across multiple seasons.
- Trained multi-class classification models with XGBoost, achieving 68.7% accuracy and a 0.65 macro F1-score.

EDUCATION

Master of Science in Data Science and Business Analytics

03/2026

Asia Pacific University of Technology & Innovation

Kuala Lumpur, Malaysia

GPA: 3.62 / 4.00

Bachelor of Science in Mathematics

04/2024

Kottakkal Farook Arts and Science College (University of Calicut)

Malappuram, Kerala, India

CERTIFICATES

TIBCO & SAS Professional Certificates in Data Science & Business Analytics (APU): Completed university-integrated professional certifications in data science and business analytics.

Guinness World Record Participant & Showcase Recipient – Kanz AI Training Hackathon: Featured Showcase Recognition (Built AI Leftover Food Assistant).